DATA SCIENCE & ANALYTICS

Artificial Intelligence & Machine Learning

This AI course introduces learners to the concepts, tools, and practical use cases shaping modern work. It is designed for learners who want to understand AI clearly and apply it responsibly in business, data, content, and technology contexts.

DURATION	STUDY FORMAT	TUITION
16 weeks	On-campus	NGN 537,500

What you will learn

- Apply AI tools to practical tasks
- Identify business opportunities for AI
- Build confidence for emerging technology roles

Tools and technologies

AI assistants / Prompt frameworks / Automation tools / Data and content workflows

Course curriculum

14 modules / 42 published learning topics

01 Module 1: Artificial Intelligence and Machine Learning Foundations

- AI problem types, learning paradigms, and real-world applications
- Distinguish rules, classical machine learning, deep learning, and generative AI
- Frame a business problem as a measurable AI task

02 Module 2: Python for AI

- Python syntax, functions, collections, files, environments, and notebooks
- NumPy arrays, vectorized operations, and reproducible experiments
- Write clean, testable data and modelling code

03 Module 3: Mathematics and Statistics for Machine Learning

- Descriptive statistics, probability, distributions, and sampling
- Vectors, matrices, similarity, gradients, and optimization intuition
- Interpret uncertainty, correlation, and model evidence responsibly

04 Module 4: Data Preparation and Exploration

- Load, clean, join, transform, and validate datasets with pandas
- Handle missing values, outliers, leakage, imbalance, and categorical data
- Explore relationships and communicate findings visually

05 Module 5: Supervised Learning

- Regression, classification, baselines, and feature engineering
- Train linear models, trees, ensembles, and nearest-neighbour models
- Select suitable algorithms for a problem and dataset

06 Module 6: Model Evaluation and Improvement

- Train, validation, and test splits; cross-validation and pipelines
- Accuracy, precision, recall, F1, ROC-AUC, and regression metrics
- Tune models and diagnose underfitting, overfitting, bias, and variance

07 Module 7: Unsupervised Learning

- Clustering, dimensionality reduction, anomaly detection, and segmentation
- Evaluate groups without labelled answers
- Build and explain a customer or operational segmentation project

08 Module 8: Deep Learning Foundations

- Neural networks, activation functions, loss, backpropagation, and optimizers
- Build and train models with a modern deep-learning framework
- Apply regularization and track experiments

09 Module 9: Computer Vision

- Image representation, preprocessing, convolutional networks, and transfer learning
- Classification and object-detection concepts
- Evaluate a practical image model and its failure modes

10 Module 10: Natural Language Processing

- Text cleaning, tokenization, embeddings, classification, and sequence models
- Transformer and large-language-model foundations
- Build and evaluate a text analysis application

11 Module 11: Generative AI and LLM Applications

- Prompting, structured output, retrieval-augmented generation, and tool use
- Ground models in trusted sources and measure response quality
- Add safety, privacy, cost, and latency controls

12 Module 12: Production Machine Learning

- Package models behind APIs and integrate them into applications
- Version data and models; monitor drift, quality, latency, and failures
- Reproduce, document, and maintain an ML system

13 Module 13: Responsible AI

- Fairness, explainability, privacy, security, and human oversight
- Identify misuse, harmful bias, and high-risk deployment contexts
- Create a model card and responsible-use plan

14 Module 14: AI and Machine Learning Capstone

- Define a problem, dataset, baseline, metrics, and acceptance criteria
- Build, evaluate, document, and deploy an end-to-end solution
- Present results, limitations, risks, and recommended next steps

Completion certificate

Loctech certificate of completion. Confirm assessment and completion requirements with Admissions.

Plan your next step

Contact Admissions to confirm the full syllabus, intake date, class timetable and payment options.

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